

# Greenroom — Settlement at The Crescent

*A walkthrough of everything that landed on this branch: the product features shipped on top of the upstream `greenroom-starter`, the engineering they sit on, the demo data they run against, and the deliverable docs they cite.*

All numbers sourced from the live build at commit `30ca767`. Runs against the artifact's local `libsql` file at `artifacts/api-server/data/greenroom.db` with 24 months of seeded settlement history at The Crescent.

|          |               |                 |                 |
|----------|---------------|-----------------|-----------------|
| 7        | 11            | 5               | 15              |
| APP TABS | API ENDPOINTS | CASE-STUDY DOCS | LIVE DEMO SHOWS |

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## 1. Project context

This branch is a response to the **Greenroom Applied AI PM case study — "Settlement at The Crescent."** The fork started from the public `samay-cbh/greenroom-starter` dataset (a fictional 250-cap live-music venue, 24 months of settlement history) and adds an opinionated booker-facing dashboard plus three decision engines that act on that history.

The thesis: The Crescent's settlement pain is not "we need a better spreadsheet" — it's "the deal types we sign keep producing settlements the wizard can't close cleanly." The work below is built around fixing that, in three phases the in-app product can ship without touching the contract layer.

## 2. Product features shipped

Everything below is live in this build — clickable, persisted, and audit-able. Nothing is mocked.

### Smart Switch decision engine

Converts an unsupported deal (vs, %-of-net, door) into a flat-equivalent or door-hybrid shape the in-app settle wizard can close. Booker sees a pending suggestion on the show detail page and can **Accept**, **Decline**, or open the audit blob.

Provenance is surfaced as a badge.

sgp\_engine   guarantee\_amount  
cell\_mean   door\_hybrid\_calc  
door\_dead\_pool   suppressed

### Smart Guaranteed Price

decision engine

7-step waterfall that projects a flat number from comparable cell history. Returns A / B / C / D confidence and a full `auditJson` blob (every input, every source, every winner-step decision). Anchored to the contract guarantee via `max(guarantee, percentagePayout)` ; rounded to nearest \$50.

### Improve Deal decision engine

Caps-only suggestions (expense cap, hospitality cap) on already-supported deals. Defaults are P75 of audit-derived historical flats. Does not convert deal types — that's Smart Switch's job.

### Settle wizard

Per-show `/shows/:id/settle` flow that walks gross box office → expenses → comps → recoups → `totalToArtist` with live math. Already supports flat and %-of-gross; Smart Switch widens what can be settled here.

### Insights — qualitative clusters

Per-cell (`dealType × sizeBucket`) dominant friction kind plus a top-5 cluster of recurring complaints, generated by a two-stage LLM pipeline (per-settlement summary, then per-cell theme clustering). Cached in-module with a race-safe pending promise.

### Insights — Switch-Savings panel

For each past-12-months settled vs/pn/door deal, computes the counterfactual: what would Smart Switch have suggested, what would the payout delta have been, how many recoup-thread minutes saved. Click to expand the per-deal reasoning.

### Artist profile /artists/:id

Per-artist history page: all shows at this venue, deal-type mix, settlement disputes, review-tone topics drawn from the LLM

### Settings — runtime LLM swap

Provider radio (Anthropic / OpenAI), per-provider key + model picker, persisted in a `settings` k/v table. Switching invalidates

enrichment, and a flagged-attention badge. Links back to each show detail.

the Insights cache so the next request re-clusters against the new model. Keys are write-only across the API — the GET response never returns raw key material.

### Needs Attention worklist

Four rule kinds: settled-with-no-settlement, closure-phrase-in-notes-but-status-open, disputed-recoups-but-signed, stale-disputed (>30d). Each carries human-readable evidence so the booker can act without context-switching.

### Per-show JSON export

`/shows/:id/export`

Full settlement payload plus an LLM-generated deep summary (uses the heavier `claude-sonnet-4-6` / `gpt-4o` override). Pasteable into post-mortems or contract reviews.

## 3. The seven tabs

The sidebar exposes seven pages. Each one is computed live — no static blobs.

| TAB             | ROUTE                                             | WHAT THE BOOKER SEES                                                                                                                                                                                      |
|-----------------|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Shows           | <code>/shows</code>                               | Every past or upcoming show as a card. Filter by deal type, size bucket, attention. Detail page joins artist, agent, agency, venue, deal, settlement, ticket sales, expenses, comps, recoups.             |
| Artists         | <code>/artists</code> · <code>/artists/:id</code> | Roster grid with show counts, top deal type, review-tone topics, attention badges. Profile page drills into per-artist history.                                                                           |
| Reports         | <code>/reports</code>                             | KPIs across all past shows: deal-type counts, in-app-tool usage rate, settlement status distribution, dispute rate, totals to artist / gross, comps & recoups.                                            |
| Deal Analysis   | <code>/deal-analysis</code>                       | Quantitative grids: complexity buckets, size buckets, profitability, costs, revenue by deal type, revenue by month (24mo), and the dealType × sizeBucket cross-tab with attention rate per cell.          |
| Needs Attention | <code>/needs-attention</code>                     | Worklist of past shows whose data smells off. Four rule kinds, each with explicit evidence.                                                                                                               |
| Insights        | <code>/insights</code>                            | Qualitative companion to Deal Analysis. Same cross-tab grid, but each cell shows the dominant friction kind + top-5 complaint clusters from LLM-enriched settlement notes. Plus the Switch-Savings panel. |
| Settings        | <code>/settings</code>                            |                                                                                                                                                                                                           |

| TAB | ROUTE | WHAT THE BOOKER SEES                                                                                            |
|-----|-------|-----------------------------------------------------------------------------------------------------------------|
|     |       | LLM provider/key/model picker. Persisted to the <code>settings</code> table; clears the Insights cache on save. |

## 4. Decision engines

### Smart Guaranteed Price (SGP)

The math layer. 7-step waterfall:

1. **Cell scan** — Pull all past settled deals in the same `(dealType, sizeBucket)` cell at this venue.
2. **Cell mean** — If the cell has  $\geq 8$  comparables, use its mean payout.
3. **Agent history** — Otherwise, weight by the agent's prior settlements at this venue.
4. **Genre fallback** — Otherwise, the corpus-wide genre mean.
5. **Percentage projection** — Apply the contract's %-clause to the expected gross.
6. **Winner step** — `winnerValue = max(guarantee, percentagePayout)`. Anchors to the contract guarantee so the suggested flat never underpays the signed number.
7. **Round** — `roundTo50(winnerValue)`. Note: rounds to nearest, so the suggestion can land up to \$24 below the contract guarantee on a single rounding hop — sub-\$50 noise.

**Confidence tier** (deterministic, no model):

| TIER | RULE                                                       | UI COPY                                                                  |
|------|------------------------------------------------------------|--------------------------------------------------------------------------|
| A    | <code>artistShowCount ≥ 3 AND winnerMargin &gt; 200</code> | "Strong data backing"                                                    |
| B    | <code>artistShowCount ≥ 1 OR agentShowCount ≥ 3</code>     | "Solid data backing"                                                     |
| C    | <code>agentShowCount ≥ 1 OR hasGenreData</code>            | "Genre-anchored"                                                         |
| D    | else                                                       | "Thin sample" → Smart Switch falls back to <code>guarantee_amount</code> |

### Smart Switch (the consumer of SGP)

Decides which suggestion shape to surface, based on deal type, size bucket, and SGP's confidence:

| DEAL TYPE     | BUCKET | SUGGESTION                                                                            |
|---------------|--------|---------------------------------------------------------------------------------------|
| vs / %-of-net | \$1–5K | SGP-projected flat (A/B) or conservative <code>guarantee_amount</code> fallback (C/D) |

| DEAL TYPE | BUCKET | SUGGESTION                                                                                                                         |
|-----------|--------|------------------------------------------------------------------------------------------------------------------------------------|
| door      | any    | Door hybrid: \$500 floor + 60% of pool above \$1,500 expense cap. Dead-pool case ( <code>avail ≤ floor</code> ) badged separately. |
| other     | —      | Not eligible — Improve-Deal handles caps; the rest is contract-layer work.                                                         |

### Improve-Deal (caps-only)

For deals the wizard already supports, surfaces P75–historical expense and hospitality caps as structural suggestions. No deal-type conversion — that's Smart Switch's lane. Audit (Apr 2026) showed cap suggestions are flat across gross levels at this venue, so no bucket-scaling.

### Three different floors — same word, different roles

"Floor" appears in three distinct mechanisms; they are not interchangeable:

| FLOOR                    | WHERE                   | MECHANISM                                                        |
|--------------------------|-------------------------|------------------------------------------------------------------|
| Contract-guarantee floor | SGP step 7              | <code>max(guarantee, percentagePayout)</code> in the winner step |
| Conservative fallback    | Smart Switch (tier C/D) | Literal copy of <code>deal.guaranteeAmount</code>                |
| Door hybrid floor        | Smart Switch (constant) | Fixed \$500 artist minimum regardless of walk-up                 |

**Honest rounding caveat:** Because `roundTo50` rounds to nearest (not up), the suggested flat can land up to \$24 below the contract guarantee. Documented in `greenroom-sgp-vs-flat.pdf` ; closeable with a one-line "round up when guarantee wins" fix.

## 5. Insights pipeline & Switch-Savings

### Two-stage LLM pipeline

- Per-settlement enrichment** ( `POST /api/insights/enrich` ). Every settlement with either notes, signoff text, a disputed status, or a disputed recoup is sent to the active LLM as a structured payload; the response is two new columns on the row: `positiveSummary` and `negativeSummary` (1–2 sentences each, strict JSON). Concurrency = 8 workers; idempotent.
- Per-cell clustering** ( `GET /api/insights` ). For each ( `dealType × sizeBucket` ) cell with a top friction kind, gather every flagged deal's `negativeSummary` and call the LLM once per cell to cluster into  $\leq 5$  themes ordered by count.

## Switch-Savings backend

`lib/switchSavings.ts` replays the entire Smart Switch engine against the last 12 months of settled vs/pn/door deals. Per deal: actual payout, counterfactual switch payout, money saved, time saved (estimated from notes/signoff/recoup activity), and a structured breakdown for the drill-down UI. Sorted desc, top 10 surfaced. Backed by `GET /api/insights/switch-savings`.

**Single LLM helper.** `lib/llm.ts` wraps the Anthropic and OpenAI SDKs behind `llmGenerateText` / `llmGenerateJson` / `llmIsConfigured`. Every call site (enrich, cluster, per-show export) goes through it. The Settings tab swaps providers globally with a single save.

## 6. Engineering stack & architecture

| LAYER       | CHOICE                                                                        | WHY                                                                |
|-------------|-------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Monorepo    | pnpm workspaces, Node 24, TypeScript 5.9                                      | Multi-artifact pattern (api-server, greenroom web, mockup-sandbox) |
| API         | Express 5, esbuild CJS bundle                                                 | Single-binary handoff, low ceremony                                |
| Database    | libsql/sqlite + Drizzle ORM, file-backed                                      | Artifact self-contained; data ships with the repo                  |
| Migrations  | Boot-time <code>ensureColumn</code> / <code>CREATE TABLE IF NOT EXISTS</code> | No drizzle-kit needed for this artifact's local DB                 |
| Validation  | Zod ( <code>zod/v4</code> ), <code>drizzle-zod</code>                         | Same schemas server-side & client-side                             |
| API codegen | Orval from OpenAPI spec                                                       | Type-safe React hooks for the frontend                             |
| Frontend    | Vite + React + wouter + Tailwind v4 + shadcn/ui + Recharts                    | Light, fast, no router bloat                                       |
| LLM         | Provider-agnostic helper, Anthropic + OpenAI SDKs                             | Runtime swap via Settings; only one module imports SDKs            |

### Architecture decisions worth calling out

- **Idempotent runtime migrations.** The artifact's DB is a local sqlite file, so schema changes are applied at server boot via small helpers in `db/index.ts`. No external migration tool required.
- **All LLM access goes through one helper.** `lib/llm.ts` is the only module that imports the Anthropic or OpenAI SDKs. Settings swaps providers globally with one save.
- **Settings keys are write-only across the API.** Status responses contain `{configured, source, model}` per provider — never raw key material.

- **Insights is cached in-module with a race-safe pending promise.** First request computes; concurrent requests await the same promise; cache invalidates on server restart, Settings save, or explicit clear.
- **Smart Switch owns flat conversion; Improve-Deal is caps-only.** Audit (Apr 2026) showed "%-deal → flat" is only data-safe in the cells Smart Switch already covers. Improve-Deal therefore emits only structural caps with P75-flat defaults.
- **Provenance enum on switch suggestions.** `sgp_engine` / `guarantee_amount` / `cell_mean` / `door_hybrid_calc` / `door_dead_pool` / `suppressed` — UI badges the source so bookers can demand sample-size discipline. Tier A demotes to B when the historical band is wider than \$1,000 (honest range display).

## 7. Data & demo seeding

The starter dataset is 24 months of synthetic settlement history at The Crescent. On top of it, this branch adds clean upcoming-show fixtures that exercise every decision-engine path, plus a separate seed that gives the SGP demo its full confidence-tier range.

### IDEMPOTENT SEED SCRIPTS

| SCRIPT                                         | WHAT IT SEEDS                                                                                                                                                                                                                                                     |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>scripts/seedNewDemoProposals.ts</code>   | 15 upcoming NEW DEMO shows: 6 vs (across \$1–5K, \$5–15K, \$15K+), 4 %-of-net (capped + uncapped), 3 door (across buckets), 2 %-of-gross. Every Smart Switch shape and Improve-Deal cap surface is covered.                                                       |
| <code>scripts/seedSmartSwitchDemo.ts</code>    | 2 pre-accepted Smart Switch examples on past shows (one flat, one door-hybrid) so <code>/shows?switched=1</code> has clean reference cases.                                                                                                                       |
| <code>scripts/seedSgpRangeDemo.ts</code>       | Reshapes the 5 vs/pn \$1–5K NEW DEMO shows so they span tiers <b>A</b> (Jordan Wells + 3 synthetic prior shows), <b>B via agent</b> , <b>B via artist</b> , and two <b>D</b> fallbacks — booker sees the full SGP range, not just the conservative fallback.      |
| <code>scripts/exportSeedsSnapshot.ts</code>    | Exports a portable JSON snapshot of every row this fork added on top of the upstream starter (artists, shows, deals, settlements, <code>switch_suggestions</code> , <code>guarantee_suggestions</code> ) plus a byte-for-byte copy of <code>greenroom.db</code> . |
| <code>scripts/exportCodeReviewBundle.ts</code> | Bundles the dataset + a schema/purpose guide written for an external code-review agent (Claude Code, etc.) so the prototype can be reviewed without running the app.                                                                                              |

**Idempotency contract.** Every seed script can be re-run safely — they tag their inserts ( `[NEW DEMO]` , `[SGP-DEMO-HIST]` ) and clear prior rows before re-seeding. The SGP-range seed also

busts the in-memory SGP context cache and force-regenerates each show's  
switch\_suggestions row.

## 8. Deliverable documents

Every PDF below lives in `exports/` and has a matching HTML source. Regenerate with WeasyPrint:  
`HTML('exports/<name>.html').write_pdf('exports/<name>.pdf')`.

| FILE                                              | WHAT IT IS                                                                                                                                                                        |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>greenroom-settlement-final.pdf</code>       | <b>The final memo.</b> The consolidated PM deliverable for the case study. Cites this prototype as evidence.                                                                      |
| <code>greenroom-settlement-redesign-v2.pdf</code> | <b>Settlement Redesign v2.1.</b> The opinionated three-phase argument: simplify deal types Smart Switch covers, layer Improve-Deal caps, then insurance.                          |
| <code>greenroom-supporting-document.pdf</code>    | <b>Features, APIs, and Tabs.</b> Engineering companion mapping every surface to the claim it supplies in the redesign.                                                            |
| <code>greenroom-insurance-pricing.pdf</code>      | <b>Insurance Products 1 &amp; 2 — Suggested Cost &amp; Pricing.</b> Per-show expected-cost derivation, 12-mo SGP backtest pricing table, tier ladder, bundles, sensitivity bands. |
| <code>greenroom-sgp-vs-flat.pdf</code>            | <b>SGP vs Flat — Technical reference.</b> Walks the 7-step waterfall, tier formulas, the three different "floor" mechanisms, and the rounding caveat.                             |
| <code>greenroom-project-overview.pdf</code>       | This document.                                                                                                                                                                    |

## 9. What this proves about the case study

The case-study prompt asks for a settlement redesign at a 250-cap venue. The deliverable on this branch answers in three layers stacked on top of each other:

1. **Diagnose the right problem.** Reports + Deal Analysis + Needs Attention together show the friction isn't "the wizard is slow" — it's "the wizard can't model 38% of the deal types Mariana signs." That's the redesign's opening move.
2. **Make the unsupported supportable.** Smart Switch + SGP + Improve-Deal are the in-app levers that move that 38% number down. Each one is shipped, each one is audit-able, each one has a provenance badge so the booker can interrogate it.



3. **Show the dollar-and-hour math.** The Switch-Savings panel replays the engine against 12 months of actual settlements. The Insurance Pricing brief takes the next step — what the platform charges for the residual risk the engine can't eliminate.

Each layer is wired to the next: SGP's confidence drives Smart Switch's source enum; Smart Switch's outputs feed the Switch-Savings backtest; the backtest feeds the Insurance Pricing brief's tier ladder. Pull on any thread and the rest of the argument comes with it.